

- 7 A beam of light consists of a continuous range of wavelengths from 430 to 600 nm. It passes through a cloud of cool hydrogen gas as shown in Fig. 7.1.

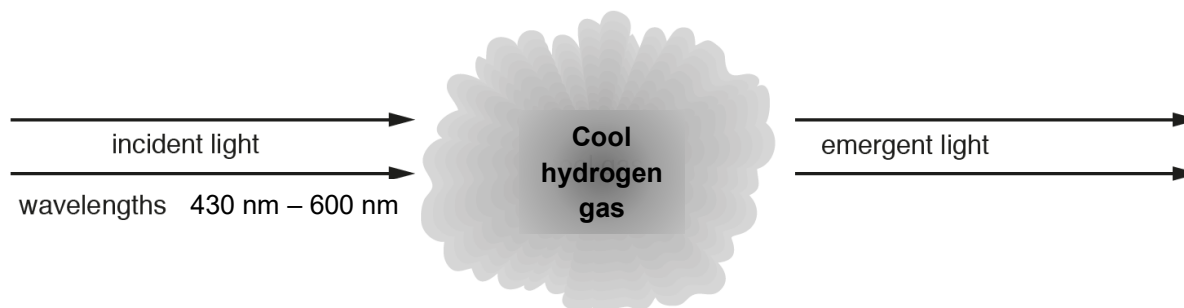


Fig. 7.1

- (a) The spectrum of light emerging from the cloud of cool hydrogen gas is viewed with a spectrometer. Explain why this spectrum contains 2 dark lines.

The incident light consist of photons of different wavelengths and different energies. A photon can only be absorbed by the cool gas and excite electron of the hydrogen atom from a lower energy level to higher energy level when the difference between the 2 energy levels are exactly equal to the energy of the photon. There are 2 possible match to the energy level differences. These 2 photons are released when the hydrogen atom undergoes de-excitation, however, these photons will be released in all direction, intensity detected will be so low that the 2 wavelength or frequent appear to be dark line.

..... [4]

- (b) The measured wavelengths, λ_m , of the selected lines in the hydrogen spectrum are given empirically by

$$\frac{1}{\lambda_m} = R \left(\frac{1}{4} - \frac{1}{m^2} \right)$$

where R is a constant of value $1.097 \times 10^7 \text{ m s}^{-1}$ and m is an integer taking the values 3, 4, 5... etc.

- (i) Calculate the value of the wavelength when $m = 4$.

$$\begin{aligned} \frac{1}{\lambda_4} &= (1.097 \times 10^7) \left(\frac{1}{4} - \frac{1}{4^2} \right) \\ \lambda_4 &= 4.86 \times 10^{-7} \text{ m} \end{aligned}$$

wavelength = m [2]

- (ii) Determine the minimum wavelength given by the equation.

$$\begin{aligned} \text{For minimum wavelength, } m \rightarrow \text{infinity, } \frac{1}{\lambda_{\min}} &= (1.097 \times 10^7) \left(\frac{1}{4} - 0 \right) \\ \lambda_{\min} &= 3.65 \times 10^{-7} \text{ m} \end{aligned}$$

- (iii) Some of the electron energy levels of the hydrogen atom in the gas are represented in Fig. 7.2.

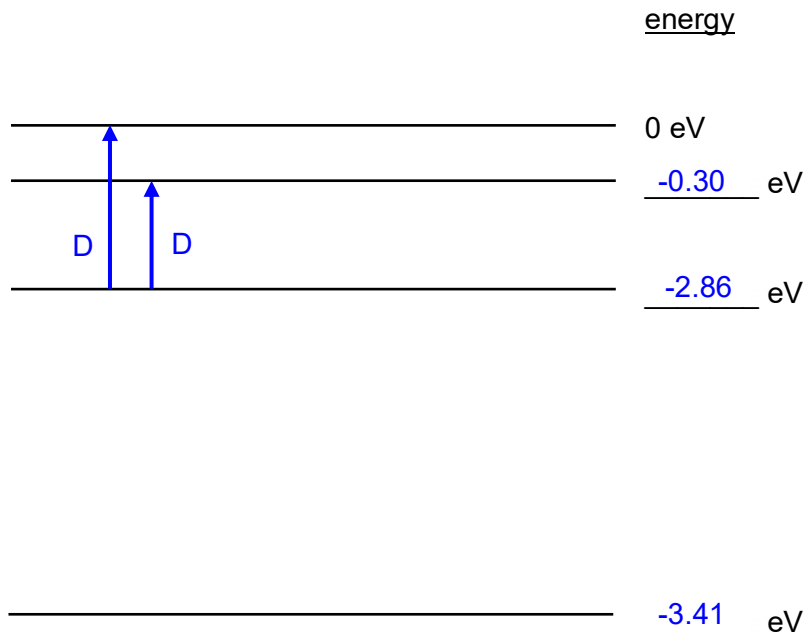


Fig. 7.2

1. Use your answers from **(b)(i)** and **(b)(ii)** to determine the energy of the 3 other electron energy levels in Fig. 7.2. Write your answer in the blanks in Fig. 7.2. [2]
2. On Fig. 7.2, draw, and label with D, the transitions giving rise to the 2 dark lines from **(a)**. [2]

[Total: 12]

$$E = h \frac{c}{\lambda_4}$$

$$E = (6.63 \times 10^{-34}) \frac{3.00 \times 10^8}{4.86 \times 10^{-7}}$$

$$\Delta E = 2.56 \text{ eV}$$

$$E = (6.63 \times 10^{-34}) \frac{3.00 \times 10^8}{3.65 \times 10^{-7}}$$

$$\Delta E = 3.41 \text{ eV}$$

$$\lambda_5 = 4.34 \times 10^{-7} \text{ m}$$

$$\Delta E = 2.86 \text{ eV}$$